

$$\vec{v}(P \in S_1 / R_0) = \vec{v}(I_1 / R_0) \quad (I)$$

$$\vec{v}(I_1 / R_0) = \frac{d}{dt} (\vec{v}(I_1 / R_0)) \Big|_{R_0}$$

$$\vec{\omega}(R' / R) = \dot{\varphi} \vec{e}_3, \quad \vec{\omega}(R_0 / R') = -\dot{\theta} \vec{u}$$

$$\vec{\omega}(R_0 / R) = \dot{\varphi} \vec{e}_3 - \dot{\theta} \vec{u}$$

$$\vec{v}(G / R_0) = \frac{d}{dt} (R \vec{u}) \Big|_{R_0} = R \dot{\varphi} \vec{u}$$

$$\vec{a}(G / R_0) = \frac{d}{dt} (R \dot{\varphi} \vec{u}) = R (\ddot{\varphi} \vec{u} + \dot{\varphi}^2 \vec{u})$$

$$\vec{v}(I_1 / R_0) = ? \quad \text{Note: } \left(\frac{d}{dt} f(x, y) \right) (x_0, y_0) \neq \left(\frac{d}{dt} f(x, y) \right) (x_0, y_0)$$

$$\text{Méthode: } \vec{v}(I_1 / R_0) = \vec{v}(G / R_0) + \vec{IG} \wedge \vec{\omega}(R_0 / R) = R \dot{\varphi} \vec{u} + r \vec{e}_1 (\dot{\varphi} \vec{e}_3 - \dot{\theta} \vec{u})$$

$$\vec{v}(I_1 / R_0) = (R \dot{\varphi} - r \dot{\theta}) \vec{u}$$

$$\text{Méthode: } \vec{v}(I_1 / R_0) = \frac{d}{dt} (\vec{HM}) \Big|_{R_0} \text{ or pour } M \in (C), \vec{HM} = \vec{HG} + \vec{GM} = R \vec{u} + r \vec{e}_1$$

$$\text{donc } \frac{d}{dt} (\vec{HM}) \Big|_{R_0} = R \dot{\varphi} \vec{u} + r (\dot{\varphi} \sin \theta \vec{u} - \dot{\theta} \vec{y}_1)$$

$$\text{or } I_0 = H | \theta = \pi \text{ aussi } \vec{v}(I_1 / R_0) = R \dot{\varphi} \vec{u} - r \dot{\theta} \vec{y}_1$$

$$\text{et } \vec{y}_1(0 = \pi) = -\vec{u} \text{ donc } \vec{v}(I_1 / R_0) = (R \dot{\varphi} - r \dot{\theta}) \vec{u}$$

$$\vec{a}(I_1 / R_0) = ?$$

$$\text{Méthode: } \vec{a}(I_1 / R_0) = \frac{d}{dt} \vec{v}(M / R_0) \Big|_{R_0} \quad M = I_2$$

$$\text{or } \frac{d}{dt} \vec{v}(M / R_0) \Big|_{R_0} = R (\ddot{\varphi} \vec{u} + \dot{\varphi}^2 \vec{u}) + r (-\dot{\varphi} \sin \theta + \dot{\varphi} \dot{\theta} \cos \theta) \vec{u}$$

$$- \dot{\varphi}^2 \sin \theta \vec{u} - \ddot{\theta} \vec{y}_1 - \dot{\theta} (\dot{\varphi} \cos \theta \vec{u} + \dot{\theta} \vec{y}_1)$$

$$\boxed{\theta = \pi}$$

$$(1)$$

$$\text{aussi } \vec{a}(I_1 / R_0) = \frac{d}{dt} \vec{v}(M \in (C) / R_0) \Big|_{R_0} \stackrel{M=I_2}{=} R (\ddot{\varphi} \vec{u} - \dot{\varphi}^2 \vec{u}) + r (+\dot{\varphi} \dot{\theta} \vec{u} - \ddot{\theta} \vec{y}_1 - \dot{\theta} (\dot{\varphi} \vec{u} + \dot{\theta} (-\vec{u})))$$



$$R_0 = (O, \vec{e}_1, \vec{e}_2, \vec{e}_3)$$

$$R = (O, \vec{u}, \vec{v}, \vec{w})$$

$$R_0 = (O, \vec{e}_1, \vec{e}_2, \vec{u})$$

Method:

$$\vec{v}(I_n/R_0) = \underbrace{\vec{v}(I_n/R_n)}_{\vec{0}} + \vec{v}(G/R_0) + \vec{\omega}(R_n/R_0) \wedge (\vec{x}(R_n/R_0) \wedge \vec{G}I_n) + \frac{d}{dt} \vec{x}(R_n/R_0) \Big|_{R_0} \wedge \vec{G}I_n + 2 \vec{\omega}(R_n/R_0) \wedge \vec{v}(I_n/R_n)$$

$$\text{so } \vec{v}(I_n/R_0) = \vec{v}(G/R_0) + \vec{\omega}(R_n/R_0) \wedge \vec{G}I_n = \underbrace{(R\ddot{\varphi} - r\ddot{\theta})}_{\vec{0}} \vec{r}$$

$$= R(\ddot{\varphi} \vec{r} - \dot{\varphi}^2 \vec{u}) + (R\dot{\varphi} \vec{e} - \dot{\theta} \vec{u}) \wedge (-r \dot{\theta} \vec{u})$$

$$+ r \vec{e} \wedge (\ddot{\varphi} \vec{e} - \ddot{\theta} \vec{u} - \dot{\theta} \dot{\varphi} \vec{u})$$

$$= R(\ddot{\varphi} \vec{r} - \dot{\varphi}^2 \vec{u}) + r \dot{\theta} (\dot{\varphi} \vec{u} + \dot{\theta} \vec{e}) + r(-\ddot{\theta} \vec{u} + \dot{\theta} \dot{\varphi} \vec{u})$$

$$= \vec{u}(-R\dot{\varphi}^2 + 2r\dot{\theta}\dot{\varphi}) + \vec{r}(R\ddot{\varphi} - r\ddot{\theta}) + r\dot{\theta}^2 \vec{e}$$

$$\vec{v}(I_n/R_0) = \vec{u}(r\dot{\theta}\dot{\varphi}) + r\dot{\theta}^2 \vec{e} = r\dot{\theta}(\dot{\varphi} \vec{u} + \dot{\theta} \vec{e}) \quad | \quad I \in R'$$

$$\vec{v}(I_n/R_0) = \vec{0}$$

Method: $\vec{v}(I'/R_0) = \frac{d}{dt} \vec{v}(M \in R') \Big|_{R_0} \quad M \in I'$

$$\overrightarrow{OM} = M\vec{u}$$

$$\frac{d}{dt} \vec{v}(M \in R') = \frac{d}{dt} (M\vec{u}) \Big|_{R_0} = \ddot{M}\vec{u} + M\dot{\varphi}\vec{e}$$

pour $M \in I'$, $\ddot{M} = 0$
d'où

$$\text{d'où } \vec{v}(I'/R_0) = R$$

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II

$I_n \in (U_n)$
 $I_a \in (U_n)$

$R_0 = (0, \vec{m}, \vec{y}, \vec{z})$
 $R' = (0, \vec{0}, \vec{v}, \vec{z})$ $\downarrow \psi$

$R_2 = (0, \vec{m}, \vec{y}, \vec{z})$
 $\xrightarrow{\psi}$

$\vec{\omega}(R'/R) = \dot{\psi} \vec{z}, \vec{\omega}(R_1/R') = \dot{\theta} \vec{m} \wedge \vec{y}$

$\vec{\omega}(R_1/R) = \dot{\psi} \vec{z} - \dot{\theta} \vec{v} = -\dot{\theta} \vec{v}$

$\vec{v}(G/R_0) = \dot{x} \vec{u} + x \dot{\psi} \vec{v}, \vec{v}(G/R_0) = \dot{x} \vec{u} + x \dot{\psi} \vec{v}$
 $+ (\dot{x} \dot{\psi} + x \ddot{\psi}) \vec{v}$
 $+ (x \dot{\psi}) \dot{\psi} \vec{v}$

$\vec{a}(G/R_0) = \vec{u}(\ddot{x} - x \dot{\psi}^2) + \vec{v}(2\dot{x} \dot{\psi} + x \ddot{\psi})$

$\vec{v}(I_n/R_0)$

- Method 1 $\vec{v}(I_n/R_0) = \frac{d}{dt} \vec{OM} \Big|_{R_0}^{M=I_n}$ ou $\vec{OM} = r \vec{m}$
 $= \dot{x} \vec{u} + x \dot{\psi} \vec{v} + r(\dot{\psi} \vec{v} + \dot{\theta} \vec{z})$

pour $M=I_n, \theta = -\pi/2, \vec{m} = -\vec{z}, \vec{y} = \vec{v}$

d'où $\vec{v}(I_n/R_0) = \dot{x} \vec{u} + x \dot{\psi} \vec{v} + r(\dot{\theta} \vec{v}) = \dot{x} \vec{u} + (x \dot{\psi} + r \dot{\theta}) \vec{v}$
 $= (\dot{x} + r \dot{\theta}) \vec{u} + x \dot{\psi} \vec{v}$

- Method 2: $\vec{v}(I_n/R_0) = \dot{x} \vec{u} + x \dot{\psi} \vec{v} + r \vec{z} \wedge (\dot{\psi} \vec{z} - \dot{\theta} \vec{v})$
 $= (\dot{x} + r \dot{\theta}) \vec{u} + x \dot{\psi} \vec{v}$

$\vec{v}(I_a/R_0)$

- Method: $\vec{v}(I_a/R_0) = \frac{d}{dt} (\vec{OM}) \Big|_{R_0}^{M=I_a}$ ou $\vec{OM} = x' \vec{u}$

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donc $\vec{v}(I_2/R) = \dot{x}\vec{u} + x\dot{\varphi}\vec{v}$. Pour $M \in I_1$, $\dot{x} = 0$, $\dot{\varphi} = 0$

$$\vec{v}(I_2/R) = \dot{x}\vec{u} + x\dot{\varphi}\vec{v} \text{ . Pour } M \in I_1, \dot{x} = 0, \dot{\varphi} = 0$$

donc $\vec{v}(I_2/R) = x\dot{\varphi}\vec{v}$

puisque: $\vec{v}(I_2/R) + \varphi\vec{\omega} \wedge x\vec{u} = \varphi x\vec{v}$

$\vec{v}(I_2/S_1) = ?$

puisque: $\vec{v}(I_2/S_1) = \vec{v}(I_2/R) - \vec{v}(I_2/S_2)$

$\vec{v}(I_2/S_1) = (\dot{x} + r\dot{\theta})\vec{u}$

puisque ~~$\vec{v}(I_2/S_1) = \frac{d}{dt}(\vec{OH})|_{M \in I_1}$~~

$\vec{v}(I_2/S_1) = \frac{d}{dt}(\vec{OH})|_{M \in I_1}$

$= \dot{x}\vec{u} + \frac{d}{dt}(r\vec{u})|_{R_1} \text{ car } \vec{v}(G/R_1) = \dot{x}\vec{u}$
 $+ \cancel{r\dot{\theta}\vec{u}} + r\dot{\theta}\vec{v}$

or pour $M \in I_1$, $\vec{v} = \vec{u}$

donc $\vec{v}(I_2/S_1) = (\dot{x} + r\dot{\theta})\vec{u}$

$\vec{v}(I_2/S_1) = ?$

puisque: $\vec{v}(I_2/S_1) = \frac{d}{dt}\vec{v}(M \in S_1/S_2)|_{R_1}$

or $\frac{d}{dt}(\vec{v}(M \in S_1/S_2))|_{R_1} = \ddot{x}\vec{u} + \dot{x}\dot{\varphi}\vec{v} + r(\ddot{\theta}\vec{u} + \dot{\theta}^2\vec{v})$

or pour $M \in I_1$, $\vec{v} = \vec{u}$, $\dot{\theta} = -\dot{\varphi}$

donc $\vec{v}(I_2/S_1) = \ddot{x}\vec{u} + r(\ddot{\theta}\vec{u} + \dot{\theta}^2\vec{v})$

$= (\ddot{x} + r\ddot{\theta})\vec{u} + r\dot{\theta}^2\vec{v} = \frac{r\dot{\theta}^2}{1}\vec{v}$

Proof:

$$\begin{aligned}\vec{a}(I_n | R_2) &= \underbrace{\vec{a}(I_n | S_1)}_{\vec{0}} + \underbrace{\vec{a}(G | S_2)}_{\vec{0}} + \underbrace{\vec{a}(S_1 | S_2)}_{\vec{0}} | R_1 \\ &\quad + \underbrace{\vec{a}(I_0 | S_2)}_{\vec{0}} + \underbrace{2\vec{a}(I_0 | S_2)}_{\vec{0}} \\ &= \ddot{m}\vec{u} + r\vec{e}_1(-\ddot{\theta}\vec{e}_2) + -\ddot{\theta}\vec{e}_2(-\ddot{\theta}\vec{e}_1 + r\ddot{\theta}\vec{e}_1) \\ &= \ddot{m}\vec{u} + r\ddot{\theta}\vec{e}_1 - \ddot{\theta}^2 r\vec{e}_2 \\ &= \ddot{m}\vec{u} + r\ddot{\theta}\vec{e}_1 - \ddot{\theta}^2 r\vec{e}_2\end{aligned}$$

$$\vec{a}(I_n | R_1) = r\ddot{\theta}\vec{e}_1 = \ddot{m}\vec{e}_1$$

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$\vec{a}(I_n | R_1)$

Proof: $\vec{a}(I_n | R_1) = \vec{a}(M \in R_1 | R_1)_{M=I_n} = \frac{d}{dt} \vec{v}(M \in R_1 | R_1) \Big|_{M=I_n}$

$$\begin{aligned}\frac{d}{dt} \vec{v}(M \in R_1 | R_1) &= \ddot{m}\vec{u} + \dot{m}(\dot{\psi}\vec{e}_1 + \dot{\theta}\vec{e}_2) \\ &\quad + (\ddot{m}\psi + m\ddot{\psi})\vec{e}_1 + m\ddot{\theta}\vec{e}_2 + r((\ddot{\psi}\cos\theta - \dot{\psi}\dot{\theta}\sin\theta)\vec{e}_1 \\ &\quad + \dot{\psi}^2\cos\theta\vec{e}_2 + \ddot{\theta}\vec{e}_1 + \dot{\theta}(\dot{\psi}\sin\theta\vec{e}_1 - \dot{\theta}\vec{e}_2)) \\ \text{for } M=I_n, \vec{a}(I_n | R_1) &= (\ddot{m}\psi + m\ddot{\psi})\vec{e}_1 - m\ddot{\theta}\vec{e}_2 \\ &\quad + r(\ddot{\theta}\vec{e}_1 + \ddot{\theta}\vec{e}_2 + \dot{\theta}(\dot{\psi}\vec{e}_1 + \dot{\theta}\vec{e}_2)) + \ddot{m}\vec{u} + \dot{m}(\dot{\psi}\vec{e}_1 + \dot{\theta}\vec{e}_2) \\ &= \vec{u}(-m\ddot{\psi}^2 + r\ddot{\theta}) + \vec{e}_1(\ddot{m}\psi + m\ddot{\psi} + r\ddot{\theta}) + r\ddot{\theta}\vec{e}_2 + \ddot{m}\vec{u} \\ &= \vec{u}(-m\ddot{\psi}^2 + r\ddot{\theta}) + \vec{e}_1(\ddot{m}\psi + m\ddot{\psi}) + r\ddot{\theta}\vec{e}_2 \\ \vec{a}(I_n | R_1) &= -m\ddot{\psi}^2\vec{u} + m\ddot{\psi}\vec{e}_1 = m\ddot{\psi}^2\vec{u} + \ddot{m}\vec{e}_1\end{aligned}$$

$$\vec{I}_n|_{R_0} = \vec{v}(\underline{I}_n|_{R_0}) + \vec{v}(\vec{e}(R_0)) + \vec{\omega}(R_0|_{R_0}) \wedge \vec{e} \vec{I}_n$$

$$+ \vec{\omega} \wedge (\vec{I}_n \wedge \vec{\omega}) + 2\vec{\omega} \wedge \vec{v}(\underline{I}_n|_{R_0})$$

$$= \vec{v}(\ddot{m} - m\dot{\varphi}^2) + \vec{v}(2\dot{m}\dot{\varphi} + m\ddot{\varphi})$$

$$+ (\ddot{\varphi} \vec{e} - \dot{\varphi} \vec{\omega} + \dot{\varphi} \dot{\varphi} \vec{e}) \wedge (-r \vec{e})$$

$$+ (\dot{\varphi} \vec{e} - \dot{\varphi} \vec{e}) \wedge (\vec{v} \vec{e} + \dot{\varphi} \vec{e})$$

$$+ r \ddot{\varphi} \vec{e} + \dot{\varphi} \dot{\varphi} r \vec{e} + r \dot{\varphi} (\dot{\varphi} \vec{e} + \dot{\varphi} \vec{e})$$

$$= \vec{v}(r \ddot{\varphi} - m\dot{\varphi}^2 + r \ddot{\varphi}) + \vec{v}(2\dot{m}\dot{\varphi} + m\ddot{\varphi} + \dot{\varphi} \dot{\varphi} r + r \dot{\varphi} \dot{\varphi}) + r \dot{\varphi}^2 \vec{e}$$

$$= -m\dot{\varphi}^2 \vec{e} + m\ddot{\varphi} \vec{e} + r \dot{\varphi}^2 \vec{e}$$

$$\vec{v}(\underline{I}_n) = -m(\dot{\varphi}^2 \vec{e} + \ddot{\varphi} \vec{e}) + \dot{\varphi} m \vec{e}$$

$$\underline{R_0|_{R_0}}: \vec{v}(\underline{I}_n|_{R_0}) = \vec{v}(M \underline{R_0|_{R_0}}) \wedge \vec{e} \vec{I}_n$$

$$\vec{v}(M \underline{R_0|_{R_0}}) = \frac{d}{dt} (M \vec{e} + M \dot{\varphi} \vec{e} + r(\dot{\varphi} \cos \theta \vec{e} + \dot{\theta} \vec{e}_\perp))$$

$$= M \vec{e} + M \dot{\varphi} \vec{e} + (\dot{M} \dot{\varphi} + M \ddot{\varphi}) \vec{e} - M \dot{\varphi}^2 \vec{e} + r((\ddot{\varphi} \cos \theta$$

$$- \dot{\varphi} \dot{\theta} \sin \theta) \vec{e} - \dot{\varphi}^2 \cos \theta \vec{e} + \dot{\theta} \vec{e}_\perp + \dot{\theta}(\dot{\varphi} \sin \theta \vec{e} - \dot{\theta} M \vec{e}_\perp)$$

$$\text{or pour } M = I_n, \theta = -\frac{\pi}{2}, \vec{e}_n = -\vec{e} \text{ et } \vec{e}_\perp = \vec{e}$$

$$\text{donc } \vec{v}(\underline{I}_n|_{R_0}) = \vec{v}(\ddot{m} - m\dot{\varphi}^2 + r \ddot{\varphi}) + \vec{v}(2\dot{m}\dot{\varphi} + m\ddot{\varphi} + \dot{\varphi} \dot{\varphi} r + r \dot{\varphi}^2 \vec{e})$$

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Cor $\ddot{m} + r\ddot{\theta} = 0 \Rightarrow \ddot{m} + r\ddot{\theta} = 0$

Alors $\vec{a}(I_1/R_0) = -m\dot{\varphi}^2 \vec{u} + m\ddot{\varphi} \vec{v} + r\dot{\theta}^2 \vec{e}_r$
 $= m[-\dot{\varphi}^2 \vec{u} + \ddot{\varphi} \vec{v}] + \underbrace{r\dot{\theta}^2 \vec{e}_r}_{= \ddot{x} \vec{e}_r}$

$\vec{a}(I_2/R_0) = ?$

meth 1:

$\vec{a}(I_2/R_0) = \underbrace{\vec{a}(I_2/R_1)}_{\vec{0}} + \underbrace{\vec{\omega}(R_1/R_0)|_{R_0}}_{\vec{0}} \wedge \vec{OI_2} + \underbrace{\vec{a}(O_1/R_0)}_{\vec{0}}$
 $+ \vec{\omega} \wedge (\vec{v} \wedge \vec{OI_2}) + 2\vec{\omega} \wedge \underbrace{\vec{v}(I_2/R_1)}_{\vec{0}}$
 $= \vec{\omega} \wedge m\vec{u} + \varphi \vec{e}_3 \wedge (\vec{0} \wedge \vec{u} \wedge m\vec{u})$
 $= m\ddot{\varphi} \vec{v} + \varphi \vec{e}_3 \wedge (m\dot{\varphi} \vec{u})$
 $= m\dot{\varphi}^2 \vec{u}$

$\vec{a}(I_2/R_0) = m[-\dot{\varphi}^2 \vec{u} + \ddot{\varphi} \vec{v}]$

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Rq: $\vec{a}(I_1/R_0) = \vec{a}(I_2/R_0) + \vec{a}(I_1/S_2)$

car $\vec{a}(I \in S_1/R_0) = \vec{a}(I \in S_1/S_2) + \vec{a}(I \in S_2/R_0)$

Ce résultat est valable dans ce cas car qu'à chaque fois l'accélération de Coriolis était nulle.
 En effet de façon générale, pour un de centre O on

$\underbrace{\vec{a}(M \in R_2/R_0)}_{\text{absolue}} = \underbrace{\vec{a}(M \in R_2/R_1)}_{\text{relative}} + \underbrace{\vec{a}(M \in R_1/R_0)}_{\text{entraînement}} + \underbrace{2\vec{\omega}(R_1/R_0) \wedge \vec{v}(M \in R_2/R_1)}_{\text{Coriolis}}$
 ou $\vec{a}(M \in R_1/R_0) = \vec{a}(O_1/R_0) + \vec{\omega}(R_1/R_0) \wedge \vec{OM} + 2\vec{\omega}(R_1/R_0) \wedge \vec{v}(M \in R_1/R_0)$

$$\vec{R}_2/\vec{R}_1 = \vec{u}(\vec{R}_2/\vec{R}_1) + \vec{v}(\vec{R}_1/\vec{R}_2) = ?$$

$$R_0 = (0, \vec{x}_0, \vec{y}_0, \vec{z}_0), R_1 = (0, \vec{x}_1, \vec{y}_1, \vec{z}_1), R_2 = (0, \vec{x}_2, \vec{y}_2, \vec{z}_2)$$

$$\vec{R}_2/\vec{R}_0 =$$

$$\vec{u}(M \in R_2/\vec{R}_0) = \vec{u}(M \in R_2/\vec{R}_1) + \vec{v}(M \in R_1/\vec{R}_0) ?$$

On a:

$$\vec{u}(M \in R_2/\vec{R}_0) = \vec{u}(M \in R_2/\vec{R}_1) + \vec{u}(O_1/\vec{R}_0) + \vec{v}(R_1/\vec{R}_0) \wedge \vec{O_1 M} \\ + \vec{v}(R_1/\vec{R}_0) \wedge (\vec{v}(R_1/\vec{R}_0) \wedge \vec{O_1 M}) + 2 \vec{v}(R_1/\vec{R}_0) \wedge \vec{v}(M \in R_2/\vec{R}_1)$$

$$\vec{u}(M \in R_1/\vec{R}_0) = \vec{u}(M \in R_1/\vec{R}_1) + \vec{u}(O_1/\vec{R}_0) + \vec{v}(R_1/\vec{R}_0) \wedge \vec{O_1 M} \\ + \vec{v}(R_1/\vec{R}_0) \wedge (\vec{v}(R_1/\vec{R}_0) \wedge \vec{O_1 M}) + 2 \vec{v}(R_1/\vec{R}_0) \wedge \vec{v}(M \in R_1/\vec{R}_1)$$

$$\text{donc } \vec{u}(M \in R_2/\vec{R}_0) = \vec{u}(M \in R_2/\vec{R}_1) + \vec{u}(M \in R_1/\vec{R}_0) \\ + 2 \vec{v}(R_1/\vec{R}_0) \wedge \vec{v}(M \in R_2/\vec{R}_1)$$

Rappel: Soit 2 repères orthonomes directs R_0, R_1

- L'application $\varphi: E \rightarrow E$

$$\vec{v} \mapsto \varphi(\vec{v}) = \left. \frac{d\vec{v}}{dt} \right|_{R_0} - \left. \frac{d\vec{v}}{dt} \right|_{R_1} \text{ est linéaire,}$$

et antisymétrique car d'après le théo de Valignon,

$$\exists \vec{R} \in E / \forall \vec{v} \in E, \varphi(\vec{v}) = \vec{R} \wedge \vec{v}$$

Dans le cadre de la cinématique $\vec{R} = \vec{\omega}(R_1/R_0)$

$$\text{Ainsi } \left. \frac{d\vec{OM}}{dt} \right|_{R_0} = \left. \frac{d\vec{OM}}{dt} \right|_{R_1} + \vec{R} \wedge \vec{OM}$$

$$\text{soit } \left. \frac{d\vec{OM}}{dt} \right|_{R_0} = \left. \frac{d\vec{OM}}{dt} \right|_{R_1} + \vec{\omega}(R_1/R_0) \wedge \vec{OM}$$

$$* \vec{V}(M/R_0) = \vec{V}(M/R_1) + \underbrace{\vec{\omega}(R_1/R_0) \wedge \vec{OM}}_{\vec{V}(M \in R_1/R_0)}$$

$$\text{Pour } M \in R_1, \vec{V}(M \in R_1/R_0) = \vec{V}(M \in R_1/R_1) + \vec{V}(R_1 \in R_0/R_0)$$

$$* \vec{\omega}(R_1/R_0) = \left. \frac{d\vec{\omega}(R_1/R_0)}{dt} \right|_{R_0} \quad \boxed{\vec{R} = \vec{\omega}(R_1/R_0)}$$

$$\begin{aligned} &= \left. \frac{d(\vec{V}(M/R_1) + \vec{V}(R_1/R_0) \wedge \vec{OM})}{dt} \right|_{R_0} \\ &= \left. \frac{d\vec{V}(M/R_1)}{dt} \right|_{R_1} + \vec{\omega}(R_1/R_0) \wedge \vec{V}(M/R_1) + \left. \frac{d\vec{\omega}(R_1/R_0)}{dt} \right|_{R_0} \wedge \vec{OM} \\ &\quad + \vec{\omega}(R_1/R_0) \wedge \left(\left. \frac{d\vec{OM}}{dt} \right|_{R_1} + \vec{\omega}(R_1/R_0) \wedge \vec{OM} \right) \end{aligned}$$

$$\text{d'où } \vec{\sigma}(M/R_0) = \vec{\sigma}(M/R_1) + \vec{\omega}(R_1/R_0) \wedge \vec{OM} + \left. \frac{d\vec{\omega}(R_1/R_0)}{dt} \right|_{R_0} \wedge \vec{OM} + \vec{\omega}(R_1/R_0) \wedge (\vec{\omega}(R_1/R_0) \wedge \vec{OM}) + 2\vec{\omega}(R_1/R_0) \wedge \vec{V}(M/R_1)$$

$$\text{donc } \vec{\omega}(M/R_0) = \vec{\omega}(M/R_1) + \vec{\omega}(R_1 \in R_0/R_0) + 2\vec{\omega}(R_1/R_0) \wedge \vec{V}(M/R_1)$$

$$\left. \frac{d\vec{V}(M \in R_1/R_1)}{dt} = \vec{\omega}(M \in R_1/R_1) \right.$$

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